Yseult Héjja-Brichard

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ACADEMIC POSITION

University of Veterinary Medicine (Interdisciplinary Life Sciences)
& University of Vienna (Behavioral and Cognitive Biology) **Vienna, Austria**
Postdoctoral researcher (1.25 year) *September 2024 - Present*

Postdoctoral advisor: Leonida Fusani (VetMed & Uni Vienna).
Focus: Studying birds' courtship displays and behaviours in the light of sensory drive and perceptual biases.

Centre d'Ecologie Fonctionnelle et Evolutive **Montpellier, France**
Postdoctoral researcher *December 2023 - August 2024*

UMBC Biological Sciences **Baltimore, MD, USA**
Postdoctoral researcher *November 2021 - November 2023*
Visiting postdoctoral researcher *March 2024 - May 2024*

Centre d'Ecologie Fonctionnelle et Evolutive **Montpellier, France**
Postdoctoral researcher *November 2020 - October 2021*

Postdoctoral advisors: Tamra C. Mendelson (UMBC) and Julien P. Renoult (CEFE, CNRS).
Focus: Investigating how information theory can help inform the evolution of sexual signalling patterns in darter fishes (*Etheostoma* spp.). Collaboration between France and the US.

EDUCATION

PhD studies in Neuroscience, Cognition, and Behaviour **2020**
Université Paul Sabatier and CerCo (CNRS), Toulouse, France

Supervisor: Benoit R. Cottereau, within the Eco-3D team.
Title of the thesis: Spatial and temporal integration of binocular disparities in primates.
Functional neuroimaging, visual behaviour, comparative visual cognition (macaques and humans)
Includes a 9-month interruption in data collection due to injury (2018–2019)

Msc in Neuroscience, Cognition, and Behaviour (2nd year) **2015**
Université Paul Sabatier, Toulouse, France

Msc in Cognitive and Social Psychology (1st year) **2014**
Université de Grenoble-Alpes, Grenoble, France

Bsc in Psychology **2013**
Université de Grenoble-Alpes, France & Universität Leipzig, Germany

RESEARCH SKILLS

Computer skills

- ***Programming:*** Matlab, R, Python (NumPy, PyTorch), C# (basic knowledge), MPI (parallel computing)
- ***Web development:*** HTML, CSS, JavaScript, Quarto
- ***Version control:*** Git, Github, Gitlab
- ***Neuroimaging:*** SPM12, caret5 (software for monkey data), PRoNTo (Pattern recognition toolbox)
- ***Stimulus presentation:*** EventIDE, jsPsych, PsychToolbox

- ***Writing:*** LaTeX, RMarkdown, Office
- ***Graphics editor:*** Adobe Illustrator, Photoshop, GIMP, Inkscape
- ***Data and article sharing:*** Github, Open Science Framework, bioRxiv, EcoEvoRXiv
- ***Animal behavior:*** BORIS, DeepEthogram, ImageJ, AnimalTA
- ***Other tools:*** MEGA software (phylogenetic data), LimeSurvey (survey tool), Blender (video animation)

Data analysis

- Image analysis (pattern recognition, signal processing)
- Functional neuroimaging data (univariate, retinotopic, and multivariate pattern analyses)
- Psychophysical and behavioural data modelling
- Eye-tracking analyses

SCIENTIFIC PRODUCTION

Publications

First or last shared authorship is indicated with an asterisk. The names of supervised students are italicised.

Under review or in preparation

- **Héjja-Brichard, Y.**, Raymond, M., Cuthill, I., Mendelson, T.C., Renoult, J.P. *(in revision)*. Background-matching patterns are attractive: support for a processing bias. [Preprint]
- Alessandroni, N. et al. (including **Héjja-Brichard, Y.**)*(in revision)*. Flexible Behavior or Flexible Methods? A Cross-Taxon Review of Experimental Designs in Reversal Learning. [Preprint]
- **Héjja-Brichard, Y.***, *MacGillavry, T.**, Fusani, L. *(in prep)*. multilevel signal functionality of super-black plumage in a bird of paradise.
- Prétôt, L. et al. (including **Héjja-Brichard, Y.**)*(in prep)*. High variability in teleost cognitive performance reveals the challenges of large-scale comparative studies in fishes.

Peer-reviewed journal articles

- Million, K., **Héjja-Brichard, Y.**, *Nasir M., Tucker, S.*, Mendelson, T.C. *(accepted at Ethology Ecology & Evolution)*. Assessing the dual utility of nuptial patterns in the rainbow darter (*Etheostoma caeruleum*).
- Prétôt, L., Agrillo, C., Bluck, B.C., Cabrera-Álvarez, M.J., **Héjja-Brichard, Y.**, ... Yeager, J. (2025). ManyFishes: A big team science collaboration on fish comparative cognition. *Animal Cognition*. https://doi.org/10.1007/s10071-025-02031-3
- Zaneva, M., Coll-Martín, T., **Héjja-Brichard, Y.**, Kalandadze, T., Kis, A., . . . Zisk, A. H. (2024). Introduction to Neurodiversity: An Annotated Reading List. *eLife*. https://doi.org/10.7554/eLife.102467 [Preprint]
- **Héjja-Brichard, Y.**, Million, K., Renoult, J.P., Mendelson, T.C. (2024). Using Neural Style Transfer to study the evolution of animal signal design: A case study in an ornamented fish. *Ecological informatics*. https://doi.org/10.1016/j.ecoinf.2024.102881 [Preprint]
- **Héjja-Brichard, Y.**, Renoult, J.P., Mendelson, T.C. (2024). Comparative analysis reveals assortative mating preferences in darters independent of sympatry and sex. *Ecology and Evolution*. https://doi.org/10.1002/ece3.11498. [Preprint]
- ManyPrimates et al. (including **Héjja-Brichard, Y.**) (2022). The evolution of primate short-term memory. *Animal Behavior and Cognition*. https://doi.org/10.26451/abc.09.04.06.2022 [Preprint]
- Guiomar, N., Krol, L.R., Combrisson, E., Dubarry, A.-S., Elliott, M.A., François, C., **Héjja-Brichard, Y.**, ... Chaumon, M. (2022). Good Scientific Practice in MEEG research: Progress and Perspectives. *Neuroimage*. https://doi.org/10.1016/j.neuroimage.2022.119056. [Preprint]
- Audurier, P., **Héjja-Brichard, Y.**, De Castro, V., Kohler, P.J., Norcia, A.M., Durand, J.-B., Cottereau, B.R. (2021). Symmetry processing in the macaque visual cortex. *Cerebral Cortex*. https://doi.org/10.

1093/cercor/bhab358. [Preprint]

- De Castro, V., Smith, A.T., Beer, A.L., Leguen, C., Vayssière, N., **Héjja-Brichard, Y.**, Audurier, P., Cottereau, B.R. & Durand, J.B. (2021). Connectivity of the cingulate sulcus visual area (CSv) in macaque monkeys. *Cerebral Cortex*. https://doi.org/10.1093/cercor/bhaa301
- **Héjja-Brichard, Y.**, Rima, S., Rapha, E., Durand, J.-B. & Cottereau, B.R. (2020). Stereomotion processing in the non-human primate brain. *Cerebral Cortex, 30*(8), 4528-4543. https://doi.org/10.1093/cercor/bhaa055. [Preprint & OSF project]
- Rima, S., Cottereau, B.R., **Héjja-Brichard, Y.**, Trotter, Y. & Durand, J.B. (2020). A new visuotopic cluster in macaque posterior parietal cortex revealed by wide-field retinotopy. *Brain Structure and Function*. https://doi.org/10.1007/s00429-020-02134-2
- Chauhan, T., **Héjja-Brichard, Y.**, & Cottereau, B.R. (2020). Modelling binocular disparity processing from statistics in natural scenes. *Vision Research*, 176. https://doi.org/10.1016/j.visres.2020.07.009
- Cottereau, B.R., Smith, A.T., Rima, S., Fize, D., **Héjja-Brichard, Y.**, Renaud, L., . . . & Durand, J.-B. (2017). Processing of Egomotion-Consistent Optic Flow in the Rhesus Macaque Cortex. *Cerebral Cortex, 27*(1), 330–343. https://doi.org/10.1093/cercor/bhw412

Other publications

- Open Science Monitoring Initiative (external contributor), *Principles of Open Science Monitoring*, 2025. https://doi.org/10.5281/zenodo.15807481

Presentations

Invited talks

- **Héjja-Brichard, Y.** (2026, Feb.) From mate preferences to cognitive abilities, what can we learn from small river fishes? WasserCluster Lunz, Lunz am See, Austria.
- **Héjja-Brichard, Y.** (2025, June) Of Fishes and Birds: Computational approaches to understanding the evolution of sexual signals signals. Department of Behavioural & Cognitive Biology, University of Vienna, Austria.
- **Héjja-Brichard, Y.** (2024) In Defense of Depth: Slow Science for Sustainable Progress in Ecology and Evolution. Society for Open, Reliable, and Transparent Ecology and Evolutionary Biology.
- **Héjja-Brichard, Y.** (2024) Neural style transfer and perceptual distance to test pattern preferences in fish. AI for ecology and evolution meeting, University of Montpellier, France.
- **Héjja-Brichard, Y.** (2023) Computational approaches to understanding the evolution of sexual signal design. Department of Biology, University of Pennsylvania, USA.
- **Héjja-Brichard, Y.** (2022) AI methods to study sexual selection. Imaginecology2: machine learning for image and sound processing and analysis in ecology, Villeurbanne, France.
- **Héjja-Brichard, Y.** (2020). Rethinking our Narratives: The Challenge of Slow Science. LiveMEEG.
- **Héjja-Brichard, Y.** & Mercier M.R. (2018). Data visualisation in cognitive neuroscience: Functional neuroimaging and electrophysiology. Toulouse Data Vizualisation group, Toulouse, France.

Conference talks

- **Héjja-Brichard, Y.**, Renoult, J.P., Mendelson, T.C. (2026). Comparative analysis reveals assortative mating preferences in darters independent of sympatry and sex. Symposium on Fish Mating Systems, Brno, Czech Republic.
- Mascolo, E. & **Héjja-Brichard, Y.** (2025). *Replicards*: Teaching evolution with a card game. EvoKE 2025 Conference, Athens, Greece.
- Million, K., **Héjja-Brichard, Y.**, Nasir M., Tucker, S., Mendelson, T.C. (2023).Territorial competition

and mate choice in the rainbow darter *(Etheostoma caeruleum)*. Southeastern Fishes Council 2023 meeting, Chattanooga, TE, USA.

- **Héjja-Brichard, Y.**, Million, K., Renoult, J.P., Mendelson, T.C. (2023). Using generative artificial intelligence to test hypotheses about animal signal evolution in an ornamented fish. Evolution 2023 meeting, Albuquerque, NM, USA.
- ManyFishes (incl. **Héjja-Brichard, Y.**)(2023) Using the detour task to assess inhibitory control across the fish taxon. 30th Annual International Conference on Comparative Cognition, Melbourne Beach, FL, USA.
- Renoult, J.P. & **Héjja-Brichard, Y.** (2022) Using deep neural networks to study the evolution of visual signals. International Conference on Ecological Sciences, SFE2-GfÖ-EEF joint meeting, Metz, France.
- **Héjja-Brichard, Y.** (2021). Workshop on Research Culture based on the Café Culture Kit developed by the Wellcome Foundation. Conférence Science Ouverte Lente Durable, Nantes, France.
- Audurier, P., **Héjja-Brichard, Y.**, Kohler, P.J., Norcia, A.M., Durand, J.-B., Cottereau, B.R. (2019). Processing of rotational symmetry in the non-human primate brain. Annual meeting of the French research group GDR Vision, Marseille, France.

Conference posters

- Prétôt, L., Bluck, B.C., Cabrera-Álvarez, M.J., **Héjja-Brichard, Y.**, Santacà, M. (2025, Oct). Hold that fin! A big-team dive into fish self-control. Big Team Science Conference, online.
- MacGillavry, T., Fusani, L., **Héjja-Brichard, Y.** (2025, July). Victoria's riflebird males create optical black holes to attract females. Animal Behavior Society, Baltimore, MD, USA.
- **Héjja-Brichard, Y.**, Million, K.M., Moodie, I.M., Renoult, J.P., Mendelson, T.C. (2024, June). Sensory drive in the era of artificial intelligence: new tools for new experiments. Animal Behavior Society, London, ON, Canada.
- **Héjja-Brichard, Y.**, Raymond, M., Cuthill, I.C., Mendelson, T.C., Renoult, J.P. (2023, Nov.). Visual pattern preferences: Testing the processing bias hypothesis with background-matching stimuli to shed light on signal design evolution. Society for Neuroscience, Washington, DC.
- **Héjja-Brichard, Y.**, Renoult, J.P., Mendelson, T.C. (2022, July). Strength of preference for conspecifics in Darters (genus Etheostoma): A meta-analysis. Animal Behavior Society, San José, Costa Rica.
- Audurier, P., **Héjja-Brichard, Y.**, Kohler, P.J., Norcia, A.M., Durand, J.-B., Cottereau, B.R. (2019, Oct.). Processing of symmetry in the non-human primate brain. Society for Neuroscience, Chicago, USA.
- De Castro, V., LeGuen, C., **Héjja-Brichard, Y.**, Audurier, P., Cottereau, B.R., Durand, J.-B. (2019, Oct.). Functional and structural connectivity of the cingulate sulcus visual (CSv) area in macaque monkeys. Society for Neuroscience, Chicago, USA.
- **Héjja-Brichard, Y.**, Bruzzone, S.E.P., Rapha, E., Durand, J.-B., Cottereau, B.R. (2019, Sept.). Influence of natural statistics on depth perception. Predictive Brain Conference, Marseille, France.
- **Héjja-Brichard, Y.**, Rima, S., Rapha, E., Durand, J.-B., Cottereau, B.R. (2018, Nov.). Stereomotion processing in the non-human primate brain. Society for Neuroscience, San Diego, USA.
- Rima, S., Cottereau, B.R., **Héjja-Brichard, Y.**, Trotter, Y., Durand, J.-B. (2017, Nov.). Wide-field retinotopy reveals a new visuotopic cluster in IPS. Society for Neuroscience, Washington DC, USA.
- **Héjja-Brichard, Y.**, Rima, S., Durand, J.-B., Cottereau, B.R. (2017, Aug.). Stereomotion processing in the non-human primate brain. European Conference on Visual Perception, Berlin, Germany.
- **Héjja-Brichard, Y.**, Rima, S., Trotter, Y., Banks, M.S., Durand, J.-B., Cottereau, B.R. (2015, Oct.). Adaptation to the 3D properties of the environment in non-human primates. GDR Vision annual forum, Grenoble, France.

TEACHING & MENTORING EXPERIENCE

Guest lecturer (2024, 2025) for MSc students in Cognition, Behaviour and Neurobiology
Department of Behavioural and Cognitive Biology, University of Vienna, Austria.
Seminar: *Advanced topics in courtship behaviour and sexual selection*

Guest lecturer (2023) for Undergraduate students in Biological Sciences
Department of Biological Sciences, University of Maryland, College Park, US.
Seminar: *Using Machine Learning to investigate communication signals in fish*

Guest lecturer (2022) for Undergraduate students in Biological Sciences
Department of Biological Sciences, UMBC, US.
Seminar: *Deep Learning et al. Architecture and Applications in Biology*

Temporary teaching assistant (2019-2020) for BSc Psychology students (1st year) - 48 hrs
Department of Psychology, Université Toulouse Jean Jaurès, Toulouse, France.
Tutorial classes: *Introduction to Psychophysiology*
Responsibilities: Re-designed and delivered weekly tutorial classes to groups of 40 students on the topics of Ontogenesis, Hormones in the human body, and Introduction to Genetics.

Graduate teaching assistant (2015-2018) for BSc Biology students (2nd and 3rd years) - 193 hrs
Department of neuroscience and behavioural science, Université Paul Sabatier, Toulouse, France.
Practical works and tutorial classes: *Nervous and cerebral functions, Behavioral ecology, Neurophysiology, Neuroscience*
Responsibilities: Designed and delivered the content of two seminars to groups of 40 students; trained two new graduate teaching assistants; assessed students' reports and final exams; implemented new assessment methods. Trained groups of 20 students on experimental designs, research methodology, and statistical analyses during weekly three-hour lab sessions.

Student supervision

- Diego Monge, Master's student in Visual Computing, Technical University of Vienna, 2026
- Géza Sümegi, Master's student in Data Science, University of Vienna, Austria, 2025
- Shivansh Sharma, Bachelor's student in Computer Science & Engineering, Babu Banarasi Das University, Lucknow, India, 2025
- Abdullah Shahzad[+], Undergraduate student in Biological Sciences, UMBC, US, 2025
- Arthur Celle, Assistant Engineer, Centre d'Ecologie Fonctionnelle et Evolutive, Montpellier, France, 2025
- Nishtha Pareek, Master's student in Biology and Electronics Engineering, Birla Institute of Technology and Science, Pilani, India, 2024
- Sarah Clement, Master's student in Ecology and Evolution, University of Rennes, France, 2024
- Maxine Akunnakwe[+], Undergraduate student in Biological Sciences, UMBC, US, 2024
- Gabriella Dennis[+], Undergraduate student in Biological Sciences, UMBC, US, 2023, 2024
- Jonah E. Brenner, Master's student in Ecology & Evolution, University of St Andrews, UK, 2023
- Yoni Renous, Master's student in Evolutionary Biology (MEME), University of Montpellier, France, 2023
- Jessie Gordon, Undergraduate student in Biological Sciences, UMBC, US, 2023
- Kossi Katchekpele, Undergraduate student in Biological Sciences, UMBC, US, 2022
- Lauren Norwood, Undergraduate student in Biological Sciences, UMBC, US, 2022
- Silvia Bruzzone, Master's student in Neuroscience, University of Trieste, Italy, 2019
- Guillaume Thuéry, Master's student in Neuroscience, University of Bordeaux, France, 2018

[+] *Undergraduate research award recipient*

Peer-mentoring

- Weaving the Future of Animal Behavior (WFAB) program, hosted by the Animal Behavior Society (Coordinator: Emilia Martins, ASU). July 2022 – August 2023. Online and in-person workshops to support early career investigators in our transition to academic positions.

- Disabled in STEM (Coordinator: Alyssa Paparella). October 2023 - July 2024. Mentorship and resource sharing for early career disabled researchers.

GRANTS & FELLOWSHIPS

Research funding

- 2026-2027: Start-Up Call @VetMedUni (co-PI): *Decoding Stress in Fish: Identifying Behavioural Signatures with Machine Learning*
- 2026-2029: FWF's Principal Investigator Project (Research collaborator): *Evolution of elaborate multi-component courtship displays*
- 2025-2028: French-Austrian Joint Project (Research collaborator): *Mating and bonding decisions in a challenging environment*
- 2024-2027: Wild Animal Initiative Research Grant (Main research collaborator): *Automated measurement of breath rate from mobile phone videos as a welfare indicator in wild great tits*
- 2023-2024: Two Undergraduate Research Awards, UMBC (Co-PI): *Investigating inhibition control in fish*; *Measuring behavioral visual acuity in darter fish*
- 2020: Three-month extension to finalise my doctoral thesis, Internal funding, CerCo - CNRS
- 2015: Three-year doctoral fellowship to pursue a PhD, University of Toulouse, France

Grants and awards

- 2025: Travel grant to attend the COSMOS summer school in Tokyo, Japan
- 2025: SIPS Commendation award for the *Annotated introductory reading list for neurodiversity* paper
- 2023: Travel grant to attend the COSYNE 2023 conference in Montréal, QC, Canada
- 2022: Travel grant to attend the Brains Minds & Machines summer course in Woods Hole, MA, USA
- 2022: Travel award to attend the annual Animal Behavior Society conference in Costa Rica
- 2021: Funding to organise a conference on open and sustainable science, LPPL and PROGEDO Loire

PROFESSIONAL AFFILIATIONS

- International Society for Behavioral Ecology (2025 - present)
- Research Network Data Science at Uni Vienna (2024 - present)
- Association for Women in Science (2023 - present)
- Animal Behavior Society (2022 - present)
- Comparative Cognition Society (2022 - present)
- Society for Open, Reliable, and Transparent Ecology and Evolutionary Biology (2021 - present)
- Association for the Study of Animal Behaviour (2020 - present)
- Big Team Science: ManyFishes (founding team member); ManyManys

EXTRACURRICULAR ACTIVITIES

Lab & Community Involvement

- Coordinating the Fusani team's participation to the Long Night of Research 2026
- Founding member of the Inclusivity working group of the WoBio initiative (University of Vienna): co-organisation of workshops and reading circles (2025)
- Statistical advisor for members of the Fusani research group (2025)
- Article for ecrLife, an eLife-supported programme to give voice to early-career researchers (2024, March): Y. Héjja-Brichard, A. Arora & R. Bajaj "Overcoming hardships for internationals in academia"
- Statistical advisor for students of the Mendelson lab (2022, 2023)
- Co-organisation of a conference to promote better scientific practices for the French-speaking community: SOLD21 (July 2021)

- Ambassador for the Center for Open Science: COS (2018-present)
- Co-creation of a Welcome Kit for new students and postdocs (2019)
- Organisation of the Lab Student Day (2019) and the Lab Winter School (2017, 2018)
- Student representative at the Lab Council (2017-2018)

Reviewing activity

- Scientific journals: *Behavioral Ecology, Ecology and Evolution, Trends in Ecology and Evolution, NeuroImage, Symmetry*
- Research grants: *The Animal Behavior Society Student Research Grants, National Science Foundation (Behavioral Systems Cluster)*
- Editorial activity: *Moderator for the EcoEvoRxiv preprint server*

Personal Implication in Science Popularisation

- Science talk *"Bird vs Fish: How to get a mate?"* for the 4th Mallnitzer Days at the National park "Hohe Tauern", Mallnitz, Austria (March 2026)
- Guest lecture on Evolutionary Biology for high school students in Greater Milan, IT and development of "Replicards", an interactive card game for science teachers (January 2025).
- Scientific speed dating with groups of high school students (December 2023; February 2025) with the Declics association.
- Outreach project at UMBC (Spring and Fall 2023) to share our research on sexual selection in darters with undergraduate and high school students.
- Editor of the second issue of the magazine "Perspectives" for FRESCO, a French national organisation involved in popularising cognitive science.
- Coordination of a thematic issue on Altruism in animals for the magazine "Perspectives" (Spring 2022).
- Public presentation on visual illusions (March 2019) for the Brain Awareness Week in Toulouse, FR.
- Public presentation on the study of 3D vision in animals (February 2017), Museum de Toulouse, FR.
- Article for the Museum of Natural History in Toulouse (2017, in French) about the evolution of 3D vision in animals: Y. Héjja-Brichard & B.R. Cottereau "Evolution et vision : le vivant a de la profondeur!"

Complementary Skills and Training

- **Languages** – French (native), English (full proficiency), German (good command in speaking, level B2), Spanish (good command in speaking, level B2), Italian (basic communication skills, level B1).
- **Methodological improvement**

  Tutorials in statistical analyses: frequentist and Bayesian frameworks

  Workshops in ethics: Primate Welfare Meetings; Research in Biology and societal considerations; Ethics and scientific integrity

  Workshop on Artificial Intelligence: 'Tracing the links between Cognition, Consciousness, and AI' by CIFAR; Tutorial on Reinforcement Learning at COSYNE 2023.
- **Summer Schools**

  Computational Summer school on Modeling Social and Collective Behavior (September 2025, Tokyo, JP). Group project: "Conformist Transmission of Behaviour in Multilevel Societies."

  Center for Brains, Minds & Machines: Summer school on the problem of intelligence (Woods Hole, MA, August 2022). Personal project on multi-agent reinforcement learning.

  Neuromatch Academy: Deep Learning (August 2021, online). Group project: "Assessing the robustness against word permutations of CNNs and LSTMs in semantic similarity learning."